

Undergraduates, digital humanities, and visualization: looking back on my early visualizations of Gravity's Rainbow

Cite as: Martin Paul Eve, "Undergraduates, digital humanities, and visualization: looking back on my early visualizations of Gravity's Rainbow", *eve.gd: Martin Paul Eve*, February 13, 2016, <https://doi.org/10.59348/jt0t4-dkq75>.

Today I don't feel well. In fact, I've got some horrible virus that's confining me mostly to bed, which I hate. So, naturally, I've spent quite a lot of time looking at Twitter while I recover (and then much longer than it would usually take me to write this post on a laptop in bed).

One thing that leaped out at me on Twitter was [Nadine Muller's series of tweets](#) about how her undergraduate dissertation mutated into her first journal article, which was a REF submission. [As she puts it](#): "So next time you hear a colleague say what undergraduates do isn't research, please think again." Too right.

Anyway, this got me thinking about what I'd done and I dug out my own undergraduate dissertation, which was on Thomas Pynchon's *Gravity's Rainbow* (1973). This was probably *the book that changed my life*, literally. As in made me want to become a literature academic. In any case, I did find one or two lines – or, at least similar ideas, that made it into my own first book.

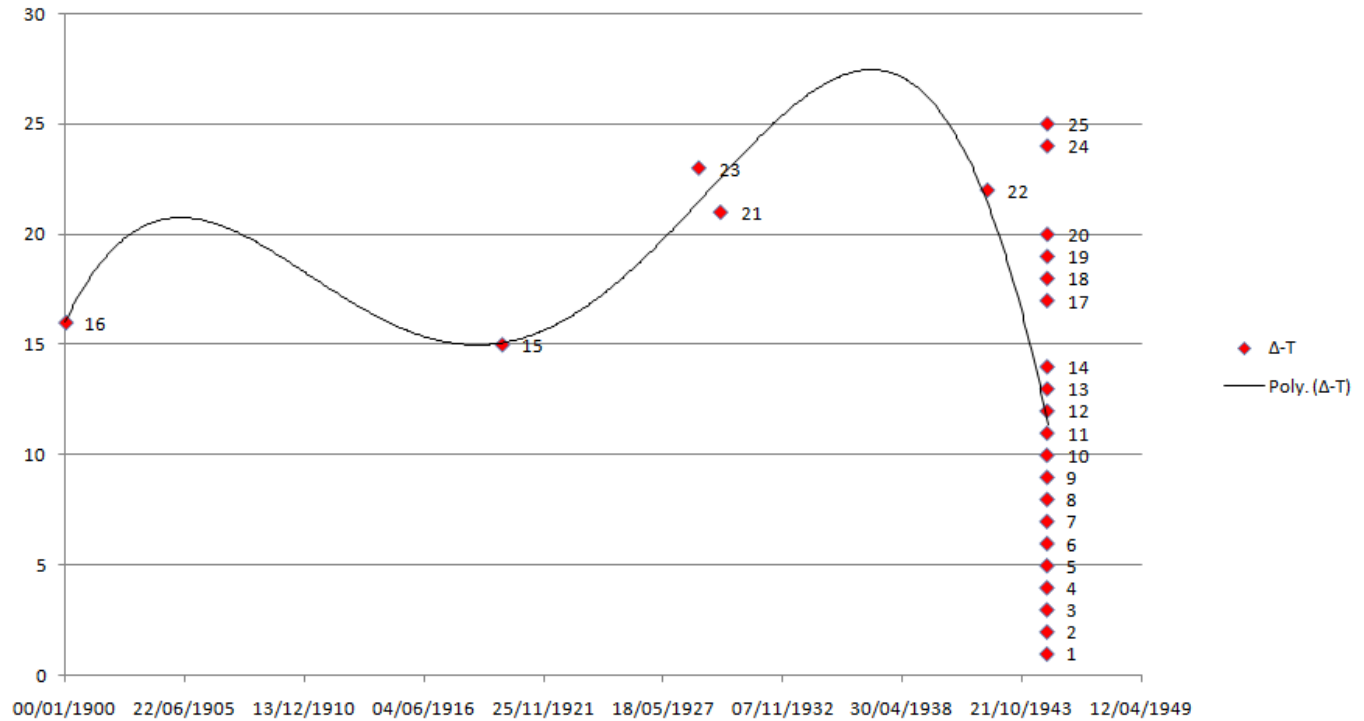
What was perhaps more interesting, though, was that in that thesis I attempted some early visualizations of the chronology of the novel. Now, nobody in the department had ever suggested to me that this was a good idea. There were no "digital humanities" faculty to show me how I could do this better or right. I hadn't ever read or seen any work that did this (so I failed the normative peer test here that I often advise for my own undergraduates when teaching). I was just interested in how discussions of time in fiction are usually framed through spatial metaphor and wanted to have a go at plotting this in some way.

Now, the results are pretty embarrassing in some ways. I was trying to push back a little against Steven Weisenburger's idea of the mandala as the overarching spatio-temporal metaphor for the text. I had an idea that the helix structure might be a better metaphor, for a host of reasons. I wrote:

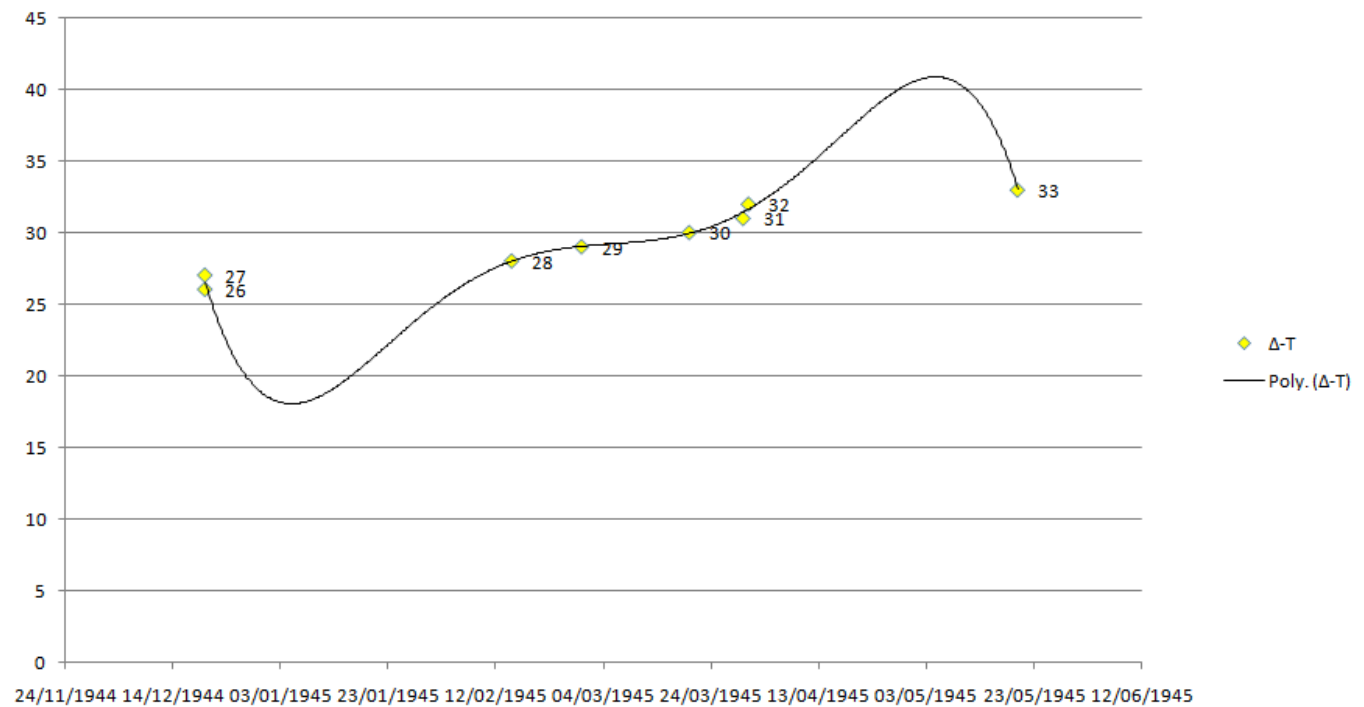
... given the early occurrence of 'living genetic chains' it is surprising that a helix, the geometry comprising half a DNA strand, an image the text clearly invokes, has not been considered as the shape of Gravity's Rainbow. This structure can be considered a superior analogy to a mandala in several respects. Firstly, when considering time as the "x-axis" of the text's already established linear historicity, both the portions above and below the helix render the parabola so frequently mentioned throughout the novel. For visual confirmation of this theory, I have situated Weisenburger's chronology in a graphical setting; see Appendix (p.32). Secondly, this structure more closely preserves the narrative's metaphorical injunction that 'It [the rocket's parabola] Begins Infinitely Below The Earth And Goes On Infinitely Back Into The Earth it's only the peak that we are allowed to see', further contributing to the already prevalent doctrine that statements about the Rocket are to be read as metatextual commentary. Thirdly, to ignore the spatio-temporal segregation of the two rockets that border the text through a circular structure is to take too literally the metaphor at the expense of its rhetorical function. As an example of this, Toni Morrison's Paradise is a novel that can be said to present a circular structure; it opens with a massacre and then relates the events leading up that event, culminating in a second narration of the exact same occurrence. In Gravity's Rainbow, on the other hand, there is no such repetition; in an apt allusion to Rilke, everything occurs 'Once, only once...'

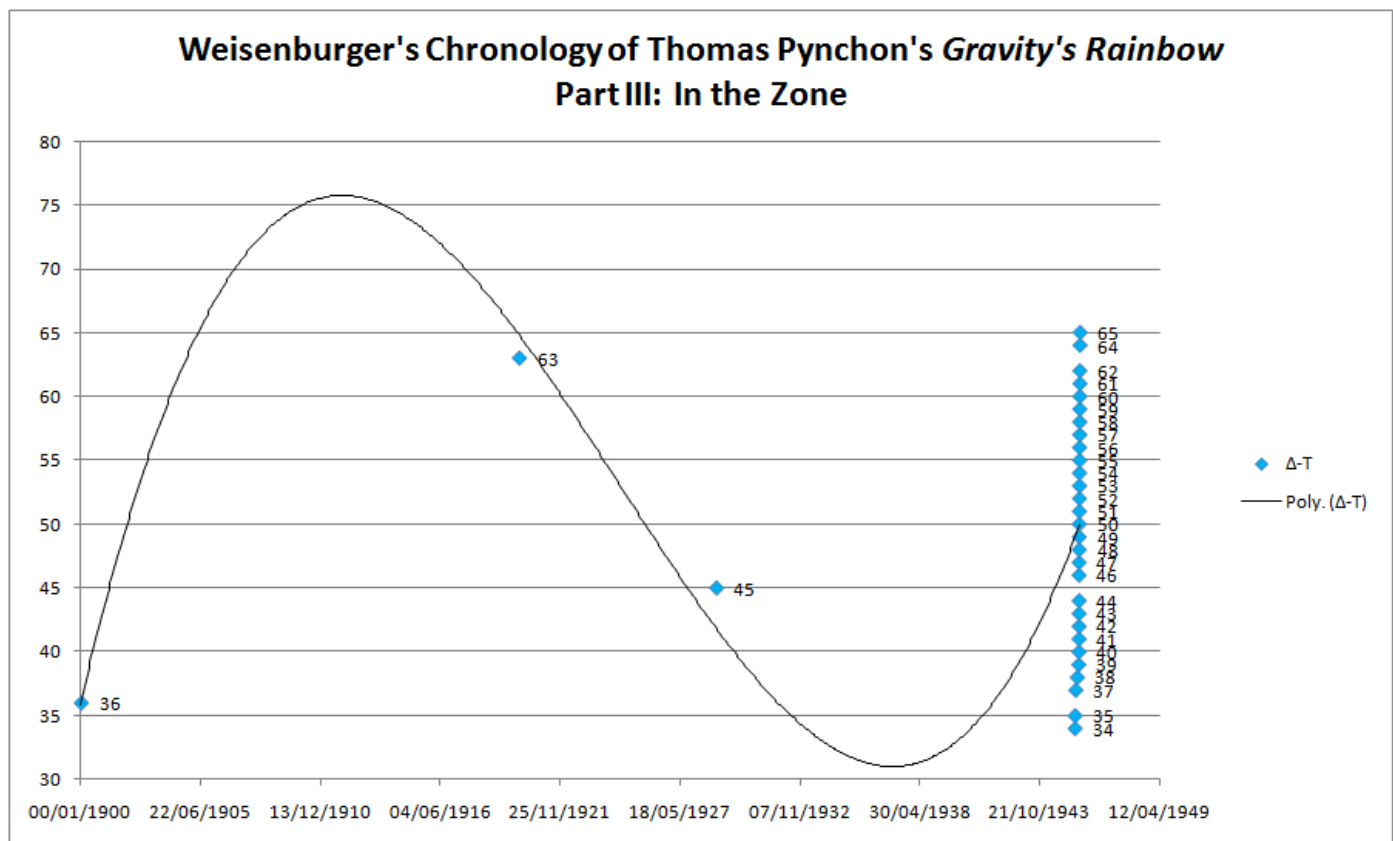
What I did, therefore, in that appendix that is mentioned, was to take the chronology that Weisenburger gives for the text, re-code it as a dataset, then plot some polynomial lines. The reason that it's a bit embarrassing is because you can probably get a polynomial line to show a helical waveform with enough parameter tweaking in the majority of cases. I'm also very unsure about the way I ordered the datapoints here. "I was young, I was foolish, I needed the dissertation" etc. But anyway, as an appendix – and not relying heavily upon this in my dissertation argument, which was mostly a close-reading exercise – I presented the following tentative graphs:

Weisenburger's Chronology of Thomas Pynchon's *Gravity's Rainbow* Part I: Beyond the Zero



Weisenburger's Chronology of Thomas Pynchon's *Gravity's Rainbow* Part II: Un Perm' au Casino Hermann Goering





Right, so lots of dodgy stuff going on here but it's still quite a fun piece of *deformance*. At a glance, I might not have even got all of Weisenburger's chronology right in my undergraduate work. What this is trying to do, though, is fundamentally something that is quite interesting to me. It's plotting one type of time axis against another; that which we might call, in the Russian Formalist tradition, *syuzhet* against that called *fabula*. Working in isolation and at an early stage in my reading and learning, I didn't realize this at the time, but what I've tried to do here is to divide up the novel into discrete chunks of time which are sequential *for the reader* (*syuzhet*) against the disordered historical chronology that the text can be said to represent (*fabula*). The axes are interchangeable is what I now see.

I still think there might be something useful to present here. It's certainly not covered by the work I did when I was an undergraduate, which I think was a graceful but naive methodological failure (and valuable as a failure). But I haven't yet seen a really good visualization of (or set of methods to visualize) how texts like *Gravity's Rainbow* structure their supposed historical chronologies, let alone the intra-diegetic extra-real chronologies.